```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *left, *right;
};

struct Node* createNode(int data) {
    struct Node* newNode =
        (struct Node*)malloc(sizeof(struct Node));

    newNode->data = data;
    newNode->left = newNode->right = NULL;

    return newNode;
}

struct Node* insert(struct Node* root, int data) {
    if (root == NULL)
        return createNode(data);

    if (data < root->data)
        root->left = insert(root->left, data);
    else if (data > root->data)
        root->right = insert(root->right, data);

    return root;
}

void inorder(struct Node* root) {
    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

void preorder(struct Node* root) {
    if (root != NULL) {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
```

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    }
}

void postorder(struct Node* root) {
    if (root != NULL) {
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
    }
}

int bstSearch(struct Node* root, int key, int *count) {
    while (root != NULL) {
        (*count)++;

        if (root->data == key)
            return 1;

        if (key < root->data)
            root = root->left;
        else
            root = root->right;
    }

    return 0;
}

int linearSearch(int a[], int n, int key, int *count) {
    for (int i = 0; i < n; i++) {
        (*count)++;

        if (a[i] == key)
            return 1;
    }

    return 0;
}

int main() {
    int a[] = {45, 20, 60, 10, 30, 50, 70, 25, 55};
    int keys[] = {25, 55, 90};
    int n = 9;
    struct Node* root = NULL;
```

```
    for (int i = 0; i < n; i++)
        root = insert(root, a[i]);

    printf("Inorder: ");
    inorder(root);

    printf("\nPreorder: ");
    preorder(root);

    printf("\nPostorder: ");
    postorder(root);

    printf("\n\nSearch Results:\n");

    for (int i = 0; i < 3; i++) {
        int bstCount = 0, linearCount = 0;

        int bstFound =
            bstSearch(root, keys[i], &bstCount);

        int linearFound =
            linearSearch(a, n, keys[i], &linearCount);

        printf("\nKey: %d\n", keys[i]);

        printf("BST Search: %s, Comparisons: %d\n",
               bstFound ? "Found" : "Not Found",
               bstCount);

        printf("Linear Search: %s, Comparisons: %d\n",
               linearFound ? "Found" : "Not Found",
               linearCount);
    }

    return 0;
}
```

```
Inorder: 10 20 25 30 45 50 55 60 70
Preorder: 45 20 10 30 25 60 50 55 70
Postorder: 10 25 30 20 55 50 70 60 45
```

Search Results:

Key: 25
BST Search: Found, Comparisons: 4
Linear Search: Found, Comparisons: 8

Key: 55
BST Search: Found, Comparisons: 4
Linear Search: Found, Comparisons: 9

Key: 90
BST Search: Not Found, Comparisons: 3
Linear Search: Not Found, Comparisons: 9